Optics (PHY5CJ302 / APH5CJ302): November 2026 paper traced to the textbooks

Paper: Fifth Semester CUFYUGP, November 2026, 2024 admission onwards (QP code D153249 / 796299). 2 hours, 70 marks. Section A: Q1–10, 3 marks each, ceiling 24. Section B: Q11–18, 6 marks each, ceiling 36. Section C: answer any one of Q19–20, 10 marks.
Book 1: Ghatak, *Optics*, 6th ed. (Modules 1–3). Pages are printed (chapter.page) with the PDF page of the full book in brackets. Book 2: Subrahmanyam, Brij Lal & Avadhanulu, *A Textbook of Optics*, ch. 20 (Module 4). Printed page with the PDF page of `SSC-V6/module4.pdf` in brackets (printed = PDF + 479). "Unit PDF" means the cut in `SSC-V6/STUDY-PLAN-V5/sources/sem5/phy/optics/`. "Your model" means your `optics-module-N-pyq+model.pdf` (M2 Q5 = Module 2 paper, question 5).

Summary

- **No question is copied from a Ghatak or Book 2 exercise.** The paper is built from standard theory questions and from older Calicut papers.
- **17 of the 21 parts are theory printed in the books.** Q11 and Q14 are among them, but the lab procedure is only partly printed. 4 are numericals of the same type as a book result or example (Q12, Q16, Q17a, Q17b).
- **Repeats from old Calicut papers:** Q1 = N21 Q13 / N22 Q13 (word for word, "sustained interference pattern"). Q5 ≈ N19 Q14 / N22 Q14 (colour of thin films). Q10 ≈ N23 Q20 (2019 admission, Gaussian formula). Q14 = N20 Q19 (biprism). Q15 ≈ N24 Q9 (Huygens). Q17 uses the quartz indices 1.544 / 1.553 from N20 Q28, N21 Q17 and N25 Q17, but asks for a 60° phase difference instead of a quarter-wave or half-wave plate.
- **Your model papers covered 19 of the 20 questions** (directly or as the same type). Not covered: **Q2** (why polarisation proves light is transverse; it is only inside an answer-key answer). Covered only partly: Q16 (your M3 Q37 uses a plane wave; this paper puts a slit source 20 cm from the edge), Q17b (a 60° phase plate, not λ/4 or λ/2) and Q7 ("how are they made": you have the half-wave definition only).
- **Module spread before ceilings:** M1 15, M2 28, M3 34, M4 21 marks on offer (syllabus 15 / 25 / 15 / 15). **Diffraction (Module 3) is about twice its syllabus share.** It has 3 of the 8 Section B questions and Q19.
- **Gaps in the unit PDFs:** Ghatak PDF p. 206 (printed 14.5, start of §14.3 Coherence, the basis of Q1) is in no cut: unit 6 stops at 205 and unit 7 starts at 207. The chapter problem lists (e.g. Ghatak Problems 20.8–20.10 on straight edges, PDF 342, the model for Q16) are also outside the cuts. Both are now cut: `module-2/gap-14.3-coherence.pdf` and `module-3/exercises-20-problems.pdf`.

Match key: **Exact** same wording as the book (none on this paper); Theory a definition, result or derivation printed in the text; Same type a book example, problem or formula applied to new data.

Section A (3 marks each, ceiling 24)

Q	Question	Where it is in the book	Match	In unit PDF?	Your model	Key point
1	What conditions must be satisfied to obtain a sustained interference pattern?	Ghatak §14.3 Coherence, 14.5–14.6 [206–207] (sources with a constant phase difference, footnote on incoherent sources); §14.4. Not set out as a numbered list in Ghatak. Old paper: N21 Q13, N22 Q13.	Theory	**Partly: p. 206 in no cut; p. 207 in unit 7**	M2 Q5 ✓	Coherent sources (constant phase difference), same λ, nearly equal amplitudes, narrow sources close together, screen far away, same state of polarisation.

2	Explain polarization of light. How does it prove that light is transverse?	Book 2 §20.1–20.2, pp. 480–481 [1–2].	Theory	Yes: unit 17	**Not as a question**	Restricting the vibrations to one plane. A longitudinal wave is the same in every direction perpendicular to its travel, so it cannot be polarised. Light can be (crossed polaroids or crystals cut it off), so its vibrations are transverse. Book 2 p. 481 [2].
3	How does a zone plate differ from a convex lens?	Ghatak §20.3 The Zone-Plate, 20.5 [330]: $f = r_1^2/\lambda$, eq. (20.20), foci at f, f/3, f/5 …	Theory	Yes: unit 15	M3 Q22 ✓	The zone plate works by diffraction and the lens by refraction. The zone plate has many foci and $f \propto 1/\lambda$ (red light focuses nearer, the opposite of a lens). Its image is less bright, and it has an extra phase difference.
4	Differentiate between Fresnel and Fraunhofer diffraction.	Ghatak §18.1, 18.4 [280] ("two categories"); §20.1, 20.1 [326].	Theory	Yes: unit 12 (p. 280); §20.1 (p. 326) not in unit 15	M3 Q1 ✓	Fresnel: source or screen at a finite distance, curved wavefronts, no lenses. Fraunhofer: both at infinity (lenses used), plane wavefronts, simple to analyse.
5	Why do soap bubbles and oil films show beautiful colors?	Ghatak §15.9 Colors of Thin Films, 15.18 [238].	Theory	Yes: unit 11	M2 Q17 ✓	Interference of light reflected from the top and bottom surfaces: $2nt\cos r = (m+½)\lambda$ is bright for a different λ at each thickness and angle in white light.
6	Distinguish between plane, circularly, elliptically polarized and unpolarized light.	Book 2 §20.3 and §20.5.1–20.5.4, pp. 482–487 [3–8].	Theory	Yes: unit 17	M4 (PYQ N20 Q23) ✓	The path traced by the tip of E: a line; a circle (equal amplitudes, phase π/2); an ellipse; random and changing quickly (unpolarised).
7	Describe how quarter wave and half wave plates are made.	Book 2 §20.19.1–20.19.2, pp. 515–516 [36–37], eqs. (20.49), (20.51).	Theory	Yes: unit 21	M4 Q18 (half-wave only)	A doubly refracting plate cut with its optic axis parallel to the faces. Thickness $d = \lambda/4(\mu_o-\mu_e)$ for a quarter-wave plate and $\lambda/2(\mu_o-\mu_e)$ for a half-wave plate.
8	What is optical path? Condition for constructive interference in terms of optical path difference.	Ghatak §3.1–3.2, 3.3–3.5 [56–58] (optical path = $\Sigma n_i l_i$); condition: §14.2, eq. (14.3), 14.3 [204].	Theory	Yes: units 1 and 6	M1 Q1 ✓	$L = \int n\,dl = c \times$ time. Phase difference = $(2\pi/\lambda_0)\Delta$. Constructive when $\Delta = m\lambda_0$.
9	Explain paraxial rays and the paraxial approximation.	Ghatak §4.2, 4.2 [80].	Theory	Yes: unit 2	M1 Q8 ✓	Rays close to the axis making small angles: $\sin\theta \approx \tan\theta \approx \theta$, $\cos\theta \approx 1$. All such rays from a point meet at one image point.

10	Obtain the Gaussian formula for a spherical refracting surface separating media n_1, n_2.	Ghatak §4.2.2, eq. (4.5), 4.2–4.3 [80–81]. Old paper: N23 Q20 (2019 admission).	Theory	Yes: unit 2	M1 (single spherical surface) ✓	$n_2/v - n_1/u = (n_2-n_1)/R$, from Fermat (or Snell) with the paraxial approximation and the Cartesian sign convention.

Section B (6 marks each, ceiling 36)

Q	Question	Where it is in the book	Match	In unit PDF?	Your model	Key point / answer
11	Experimental method of determining λ of sodium light using Newton's rings.	Ghatak §15.10 Newton's Rings, 15.18–15.20 [238–240], eqs. (15.63)–(15.66); travelling microscope p. 15.20 [240]. Ghatak gives the theory; the step-by-step lab procedure is general knowledge or the lab manual.	Theory	Yes: unit 11	M2 Q41 ✓	$D^2_n = 4nR\lambda$; $\lambda = (D^2_{n+p} - D^2_n)/(4pR)$. Measure the ring diameters with a travelling microscope and R with a spherometer.
12	Single slit, Fraunhofer: screen 2 m, slit 0.2 mm, first minima 5 mm either side of the centre. Find λ.	Ghatak §18.2.1, minima $b \sin\theta = m\lambda$, 18.6–18.7 [282–283]; Example 18.1, 18.7 [283].	Same type	Yes: unit 12	M3 Q38 (same type) ✓	$\lambda = b \cdot y/D = 0.2\times10^{-3} \times 5\times10^{-3} / 2 = 5\times10^{-7}$ m = **5000 Å**.
13	Explain how a zone plate behaves like a lens of variable focal length.	Ghatak §20.3, 20.5 [330], eq. (20.20) $1/v - 1/u = 1/f$ with $f = r_1^2/\lambda$; foci at f/3, f/5 ….	Theory	Yes: unit 15	M3 Q36, Q22 ✓	"Variable" means f depends on λ ($f = r_1^2/\lambda$) and that there are many foci f/(2p−1). Prove the lens formula from the zone radii.
14	What is Fresnel's biprism? How is it used to find λ?	Ghatak §14.8 Fresnel Biprism, 14.14–14.15 [215–216] (construction, virtual sources, $\beta = \lambda D/d$). Measuring d by the displacement (two-position lens) method is not in Ghatak. Old paper: N20 Q19.	Theory	Yes: unit 8	M2 Q27 (= N20 Q19) ✓	$\lambda = \beta d/D$. Measure β with the eyepiece; $d = \sqrt{(d_1 d_2)}$ from the two lens positions.
15	Discuss Huygens' theory of double refraction in uniaxial crystals.	Book 2 §20.11.1, p. 498 [19]; §20.14 Huygens' construction, pp. 504–507 [25–28].	Theory	Yes: unit 20	M4 Q15, Q41 ✓	O-wavelet is a sphere and the E-wavelet a spheroid; they touch along the optic axis. Negative crystal: oblate spheroid, outside. Positive crystal: prolate spheroid, inside. Draw the constructions for the different optic-axis orientations.

16	Slit (λ = 4500 Å) 20 cm from a straight edge; screen 60 cm from the edge. Distance between the 2nd and 3rd maxima.	Ghatak §20.6, 20.9–20.12 [334–337] (plane wave: $y = v_0\sqrt{(\lambda d/2)}$, maxima at $v_0 = 1.22, 2.34, 3.08$). Same type: Problems 20.8–20.10, 20.17 [342]. Ghatak has no point-source version.	Same type	**Theory yes (unit 16); problems page 342 no**	M3 Q37 (plane wave only)	Point source: $y = v\sqrt{(\lambda b(a+b)/2a)} = v \times 0.735$ mm ($a = 0.2$ m, $b = 0.6$ m). $\Delta y = (3.08-2.34) \times 0.735 \approx$ **0.54 mm**. The half-period-zone formula $x_n = \sqrt{((2n-1)\lambda b(a+b)/a)}$ gives 1.80 and 2.32 mm, $\Delta \approx$ **0.52 mm**; either method should get the marks.
17a	Least thickness of a calcite plate converting plane- to circularly-polarised light; $\mu_o = 1.658$, $\mu_e = 1.486$, $\lambda = 5890$ Å. (3)	Book 2 §20.19.1, eq. (20.49), p. 516 [37]; §20.21 (circular light needs a λ/4 plate at 45°), p. 518 [39].	Same type	Yes: units 21–22	M4 quarter-wave sums ✓	$d = \lambda/4(\mu_o-\mu_e) = 5890\times10^{-10} / (4 \times 0.172)$ = **8.56×10^{-7} m ≈ 0.856 µm**.
17b	Quartz, faces ∥ optic axis, λ = 5893 Å, $\mu_o = 1.544$, $\mu_e = 1.553$: thickness for a 60° phase difference. (3)	Book 2 §20.17 Phase difference between e-ray and o-ray (path difference $(\mu_e-\mu_o)d$, phase = $2\pi/\lambda \times$ path), pp. 511–512 [32–33]. Same indices as N20 Q28, N21 Q17, N25 Q17.	Same type	Yes: unit 21	M4 (λ/4, λ/2 only)	$\pi/3 = (2\pi/\lambda)(0.009)d \Rightarrow d = \lambda/(6 \times 0.009)$ = **1.09×10^{-5} m ≈ 10.9 µm**.
18	Explain the laws of refraction on the basis of Fermat's principle.	Ghatak §3.2, 3.5–3.6 [58–59] (Snell's law from minimum time).	Theory	Yes: unit 1	M1 Q2 and derivation ✓	$t = n_1\sqrt{(h_1^2+x^2)}/c + n_2\sqrt{(h_2^2+(L-x)^2)}/c$; $dt/dx = 0 \Rightarrow n_1 \sin i = n_2 \sin r$. The rays are coplanar.

Section C (answer any one, 10 marks)

Q	Question	Where it is in the book	Match	In unit PDF?	Your model	Key point
19	Discuss the theory of formation of fringes in Fresnel diffraction at a straight edge.	Ghatak §20.6, 20.9–20.12 [334–337]: Fresnel integrals, Table 20.1, Cornu spiral, eqs. (20.42)–(20.62).	Theory	Yes: unit 16	M3 Q42 ✓	$I = \frac{1}{2}I_0[(\frac{1}{2}+C(v_0))^2 + (\frac{1}{2}+S(v_0))^2]$. $I = I_0/4$ at the shadow edge, falls steadily inside the shadow, and has unequal fringes outside (first maximum 1.37 I_0 at $v_0 = 1.22$). Many books use the half-period-strip method instead; either gets the marks.
20	Young's double-slit experiment with diagram; fringe width; intensity distribution; conditions for maxima and minima.	Ghatak §14.4–14.5, 14.6–14.10 [207–211], eqs. (14.20) $x_n = n\lambda D/d$ and (14.28)–(14.30), $I = 4I_0\cos^2(\delta/2)$.	Theory	Yes: unit 7	M2 Q39 ✓	$\Delta = xd/D$; bright when $\Delta = n\lambda$, dark when $\Delta = (n+\frac{1}{2})\lambda$; $\beta = \lambda D/d$; $I = 4I_0\cos^2(\pi xd/\lambda D)$.

Module spread (marks on offer, before ceilings)

Module	Questions	Marks on offer	Share	Syllabus marks (of 70)	Syllabus share
I Fermat, geometrical optics	Q8, Q9, Q10, Q18	15	15%	15	21%
II Interference	Q1, Q5, Q11, Q14, Q20	28	29%	25	36%
III Diffraction	Q3, Q4, Q12, Q13, Q16, Q19	34	35%	15	21%
IV Polarisation	Q2, Q6, Q7, Q15, Q17	21	21%	15	21%
Total		**98**	100%	**70**	

Placement note: Q8 is put in Module I (optical path, §3.1). If it is counted in Module II (its second half is the interference condition), the shares become I 12, II 31.

Where to look next time

Book section	Pages (printed [PDF])	Questions on this paper	In unit PDFs?
Ghatak §3.1–3.2 Fermat	3.3–3.6 [56–59]	Q8, Q18	Yes (unit 1)
Ghatak §4.2 single spherical surface	4.2–4.3 [80–81]	Q9, Q10	Yes (unit 2)
Ghatak §14.3 Coherence	14.5–14.6 [206–207]	Q1	**p. 206 missing**
Ghatak §14.4–14.5 Young, §14.8 biprism	14.6–14.15 [207–216]	Q20, Q14	Yes (units 7, 8)
Ghatak §15.9–15.10 thin films, Newton's rings	15.18–15.20 [238–240]	Q5, Q11	Yes (unit 11)
Ghatak §18.1–18.2 single slit	18.4–18.7 [280–283]	Q4, Q12	Yes (unit 12)
Ghatak §20.3 zone plate; §20.6 straight edge	20.5 [330]; 20.9–20.12 [334–337]	Q3, Q13, Q16, Q19	Yes (units 15, 16)
Ghatak ch. 20 Problems 20.8–20.10	20.17 [342]	Q16 (type)	**No**
Book 2 §20.1–20.5 introduction, types	480–487 [1–8]	Q2, Q6	Yes (unit 17)
Book 2 §20.11, §20.14 Huygens	498, 504–507 [19, 25–28]	Q15	Yes (unit 20)
Book 2 §20.17–20.19 phase difference, retarders	511–517 [32–38]	Q7, Q17a, Q17b	Yes (unit 21)

Lesson: this paper rewards the **standard theory questions** of each unit and the **old Calicut papers**. Your Module 1–4 model papers already held 19 of the 20 questions. Add "why polarisation proves light is transverse", a straight edge with a point source at a finite distance, and a wave plate for an arbitrary phase difference.